

Universal Survival, First Descent, and Discrete Arcsine Laws for Continuous Symmetric Random Walks

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Abstract

For a random walk with iid, continuous increments symmetric about zero, we derive the distribution-free survival probabilities $q_n = 4^{-n} \binom{2n}{n}$ from a factorization at the unique maximum. This also closes the first strict descent and the full law of the maximum time. The Sparre–Andersen permutation identity then gives the same discrete-arcsine law for the number of positive partial sums, and both statistics converge to the Beta(1/2, 1/2) law.

1 Frozen convention and theorem

Let $X_1, X_2, \dots$ be iid real random variables with a continuous law symmetric about zero, and set $S_0 = 0$ and $S_n = X_1 + \dots + X_n$. Continuity implies that $S_0, \dots, S_n$ are distinct almost surely: each difference $S_j - S_k$ has a continuous law. We therefore use strict positivity throughout and define

$$q_n = \mathbb{P}(S_1 > 0, \dots, S_n > 0), \quad q_0 = 1,$$

$$\tau^- = \inf\{n \geq 1 : S_n < 0\}, \quad N_n = \#\{1 \leq j \leq n : S_j > 0\}, \quad M_n = \arg \max_{0 \leq j \leq n} S_j.$$

The maximum time includes zero and is unique. These convention choices are part of the theorem, not cosmetic endpoint rules.

Theorem 1 (Sparre–Andersen fluctuation closure). *For every $n \geq 0$,*

$$q_n = \frac{1}{4^n} \binom{2n}{n}, \quad Q(z) := \sum_{n \geq 0} q_n z^n = (1 - z)^{-1/2}. \quad (1)$$

For $n \geq 1$,

$$\mathbb{P}(\tau^- > n) = q_n, \quad \mathbb{P}(\tau^- = n) = q_{n-1} - q_n = \frac{q_{n-1}}{2n}. \quad (2)$$

For $0 \leq k \leq n$,

$$\mathbb{P}(M_n = k) = q_k q_{n-k}. \quad (3)$$

Moreover,

$$\mathbb{P}(N_n = k) = q_k q_{n-k}, \quad (4)$$

and both N_n/n and M_n/n converge weakly to the law with density $[\pi \sqrt{x(1-x)}]^{-1} \mathbf{1}_{(0,1)}(x)$.

The fluctuation identity is classical; our primary reference is Sparre Andersen [1]. The contribution here is a source-locked complete proof and executable convention audit, not a literature-priority claim.

2 Maximum factorization forces survival

Fix $0 \leq k \leq n$. The event $M_n = k$ requires

$$S_k - S_{k-j} > 0 \quad (1 \leq j \leq k), \quad S_{k+j} - S_k < 0 \quad (1 \leq j \leq n - k).$$

The first set of inequalities is survival for the reversed iid block $X_k, \dots, X_1$ and has probability q_k . The second uses the independent block $X_{k+1}, \dots, X_n$; sign symmetry turns it into survival of length $n - k$. Hence

$$\mathbb{P}(M_n = k) = q_k q_{n-k}. \quad (5)$$

Since the unique maximum occurs somewhere, $\sum_{k=0}^n q_k q_{n-k} = 1$. Thus $Q(z)^2 = (1-z)^{-1}$. The condition $Q(0) = 1$ selects the positive formal square root, proving (1) and (3).

The event $\{\tau^- > n\}$ agrees almost surely with strict survival, so its probability is q_n . Consecutive tail differences give (2), because $q_n/q_{n-1} = (2n-1)/(2n)$. Notice that no increment density appears in this argument.

3 Positive counts and the common arcsine limit

The remaining finite-time identity follows from the permutation-cycle lemma. We record the exact form needed here.

Lemma 1 (Sparre–Andersen cycle identity). *Under the frozen iid and no-ties assumptions,*

$$1 + \sum_{n \geq 1} z^n \mathbb{E}[u^{N_n}] = \exp \left\{ \sum_{m \geq 1} \frac{z^m}{m} (\mathbb{P}(S_m < 0) + u^m \mathbb{P}(S_m > 0)) \right\}. \quad (6)$$

Proof. Condition on increments in general position and average over their permutations. Decompose a permutation into cycles. Rotate every cycle at its unique extremal partial sum: a positive-total cycle contributes its full length to the positive-partial-sum count, and a negative-total cycle contributes zero. Ordering the rotated cycles by extremal level is a bijection back to linear increment orderings. The labeled-cycle formula contributes $1/m$ for a cycle of length m . Averaging its sign over iid increments gives the two probabilities in (6). Continuity supplies the required unique rotations. $\square$

Symmetry makes both probabilities in (6) equal to $1/2$, so

$$1 + \sum_{n \geq 1} z^n \mathbb{E}[u^{N_n}] = \frac{1}{\sqrt{(1-z)(1-uz)}} = \sum_{n \geq 0} \sum_{k=0}^n q_k q_{n-k} u^k z^n. \quad (7)$$

Coefficient extraction proves (4).

Stirling's formula gives $q_m \sim (\pi m)^{-1/2}$. For $k = \lfloor nx \rfloor$ with x in a compact subset of $(0, 1)$,

$$n q_k q_{n-k} \longrightarrow \frac{1}{\pi \sqrt{x(1-x)}}.$$

The bound $q_m \leq C/\sqrt{m+1}$ makes the total mass in $k \leq \varepsilon n$ at most $C' \sqrt{\varepsilon}$, uniformly in n ; symmetry handles the other endpoint. A bulk Riemann sum followed by $\varepsilon \downarrow 0$ proves the weak limit for M_n/n , and (4) gives the same limit for N_n/n .

References

- [1] E. Sparre Andersen, *On the fluctuations of sums of random variables*, *Mathematica Scandinavica* **1** (1953), 263–285, [doi:10.7146/math.scand.a-10385](https://doi.org/10.7146/math.scand.a-10385).